

Fig. 2: An actual-size, single-side PCB for the medicine-time reminder

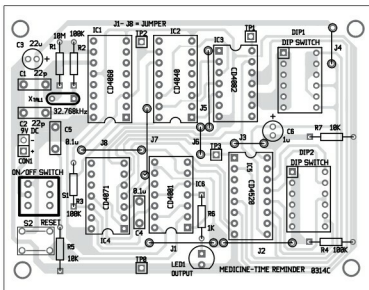


Fig. 3: Component layout for the PCB

PARTS LIST

Semiconductors:

- IC1 - CD4060 14-stage ripple carry binary counter
- IC2 - CD4040 12-stage ripple carry binary counter
- IC3 - CD4082 dual 4-input AND gate
- IC4 - CD4071 quad 2-input OR gate
- IC5 - CD4520 dual upcounter
- IC6 - CD4001 quad 2-input NOR gate

LED1 - 5mm red LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 10-mega-ohm
- R2-R4 - 100-kilo-ohm
- R5, R7 - 10-kilo-ohm
- R6 - 1-kilo-ohm

Capacitors:

- C1, C2 - 22pF ceramic disk
- C3 - 22 μ F, 16V electrolytic
- C4, C5 - 0.1 μ F ceramic disk
- C6 - 1 μ F, 16V electrolytic

Miscellaneous:

- X_{TA}1 - 32.768kHz
- CON1 - 2-pin connector
- S1 - On/off switch
- S2 - Tactile switch
- DIP1, DIP2 - 12-pin DIP switch
- 9V PP3 battery/9V, 500mA regulated supply

DIP2 switches. The positions of DIP1 and DIP2 allow selection of 4, 6, 8, 12, 24 and 48-hour time intervals. The set-reset flip-flop formed by G1 and G4 of IC6 is set through G4 of IC4 each time a low-to-high transition is there at the pin of IC5 selected by DIP2. When the flip-flop is set, gate G2 of IC6 is enabled and the 2Hz frequency available at pin 3 of IC1 becomes available at pin 4 of IC6, causing LED1 to flash. The flip-flop can then be reset by means of S2. A master reset is automatically done at power-on by means of C6 and R7. Refer Table I for selecting the time periods.

First choose the hour interval using DIP1 and then apply power through S1. After the specified interval of time has elapsed, LED1 will start flashing at 2Hz. This status will last until pushbutton S2 is pressed, but the circuit will continue the counting and LED1 will flash again when

the same hour interval as before is reached. An important feature of this circuit not found in similar devices is that the internal counter is not reset when S2 is pressed, which allows a better time-interval precision.

Construction and testing

An actual-size, single-side PCB for the medicine-time reminder circuit is shown in Fig. 2 and its component layout in Fig. 3.

After assembling the circuit on a PCB, enclose it in a suitable plastic case such that LED1, S1, S2, DIP1 and CON1 stick out. Verify voltage levels on various test points mentioned in the test point table before using the circuit. ●

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